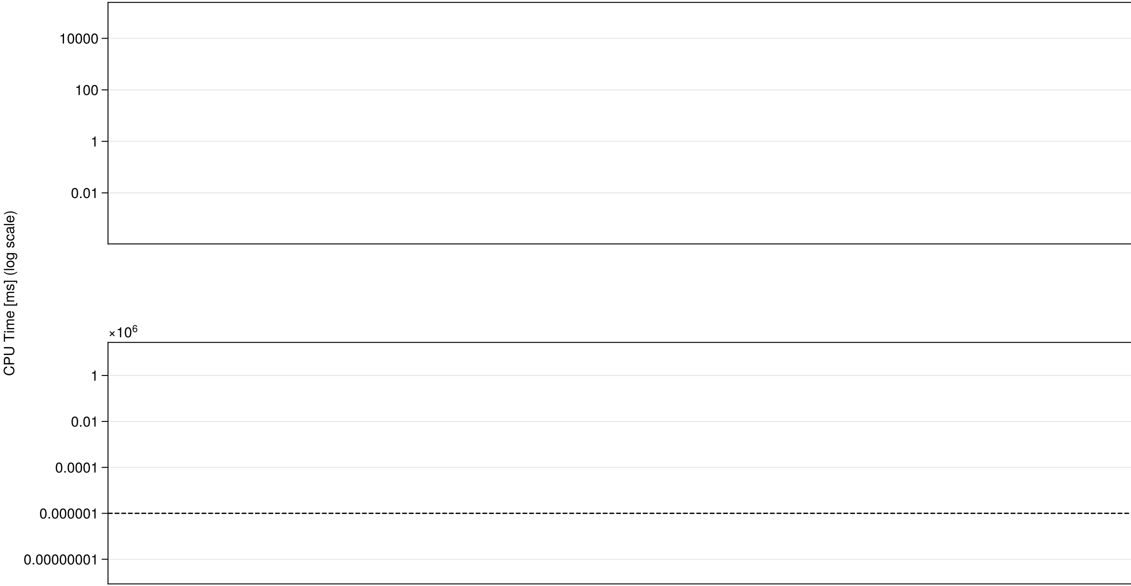
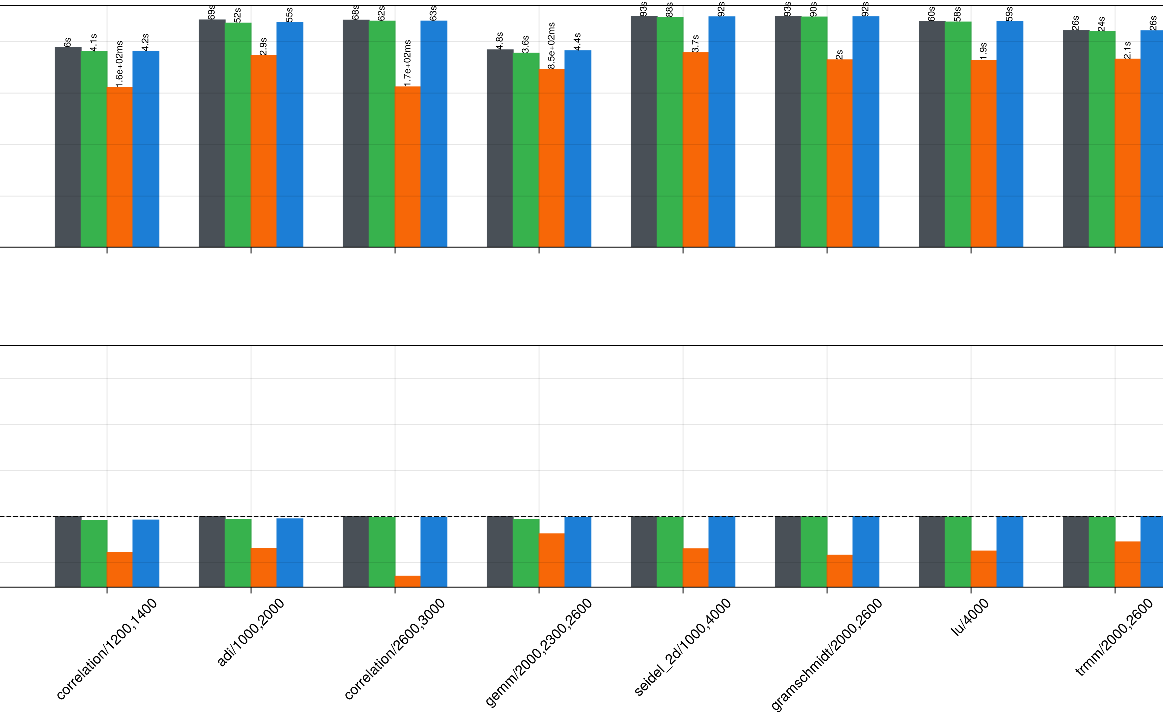
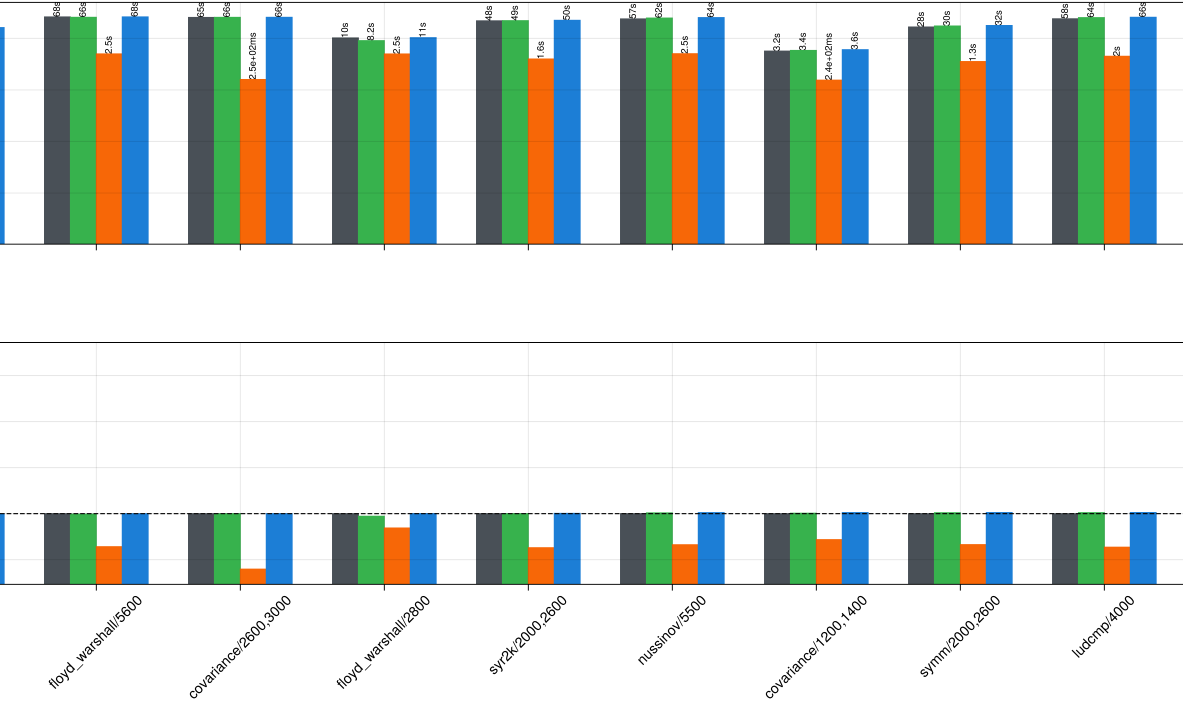
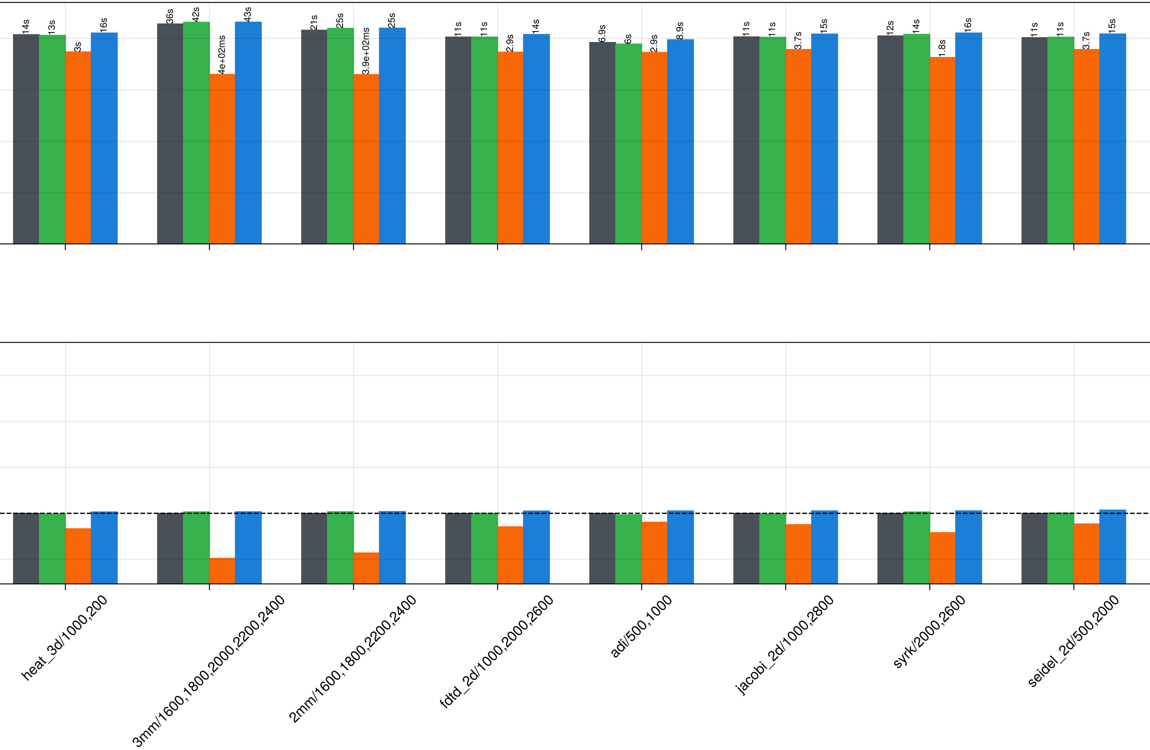
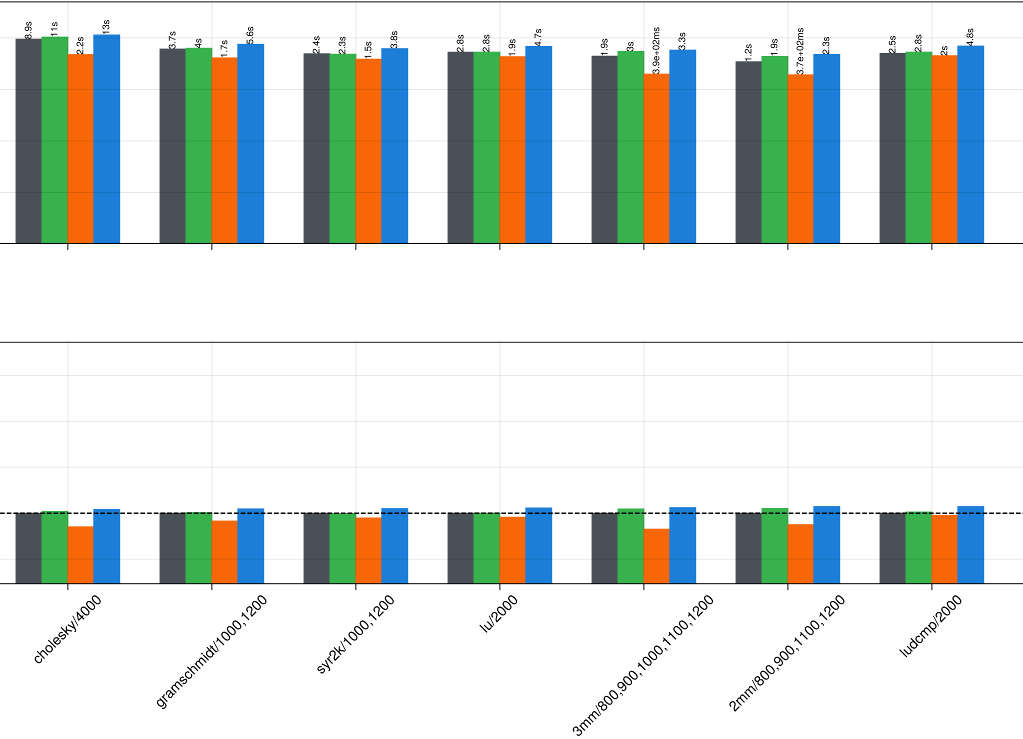
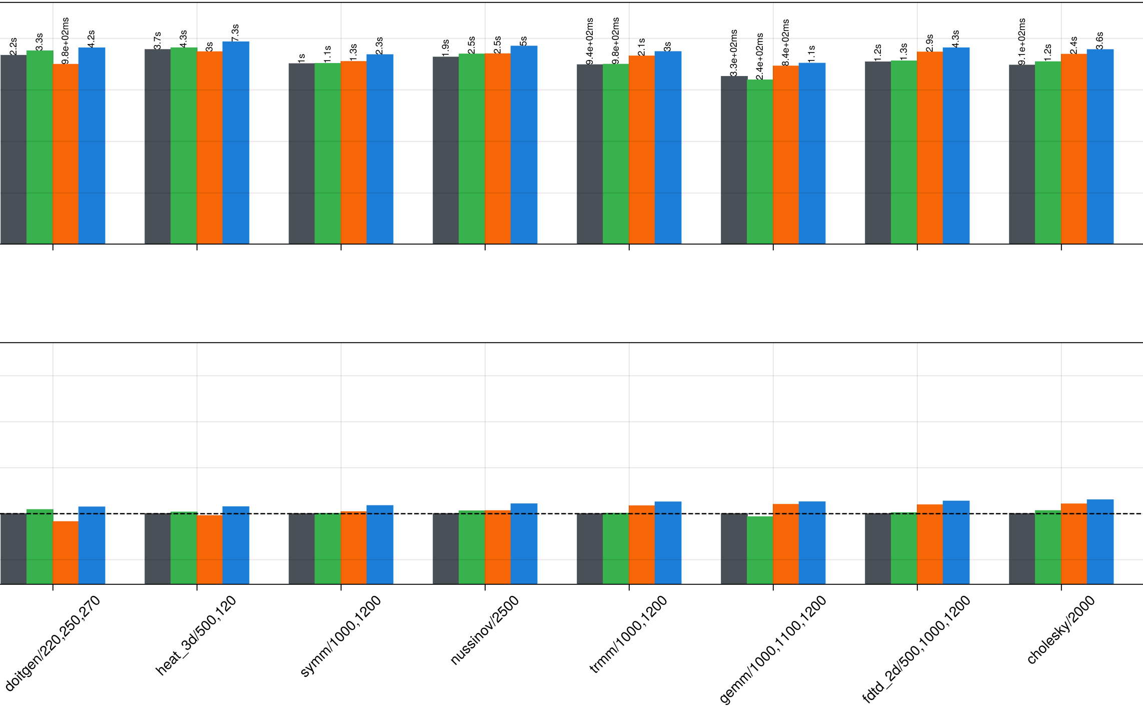

CPU Time [ms] (log scale)
10000
100
1
0.01
×10^6
1
0.01
0.0001
0.000001
0.00000001

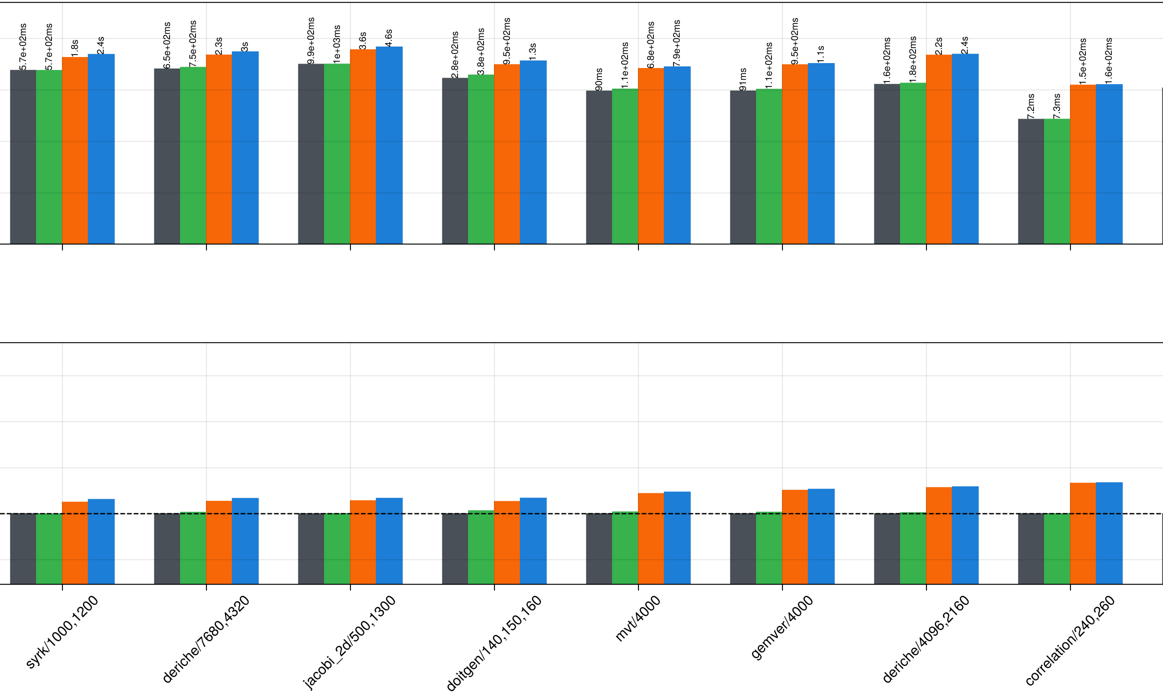

6s
4.1s
1.6e+02ms
4.2s
69s
52s
2.9s
55s
68s
62s
1.7e+02ms
63s
4.8s
3.6s
8.5e+02ms
4.4s
93s
88s
3.7s
92s
93s
90s
2s
92s
60s
58s
1.9s
59s
26s
24s
2.1s
26s
correlation/1200,1400
adi/1000,2000
correlation/2600,3000
gemm/2000,2300,2600
seidel_2d/1000,4000
gramschmidt/2000,2600
lu/4000
trmm/2000,2600

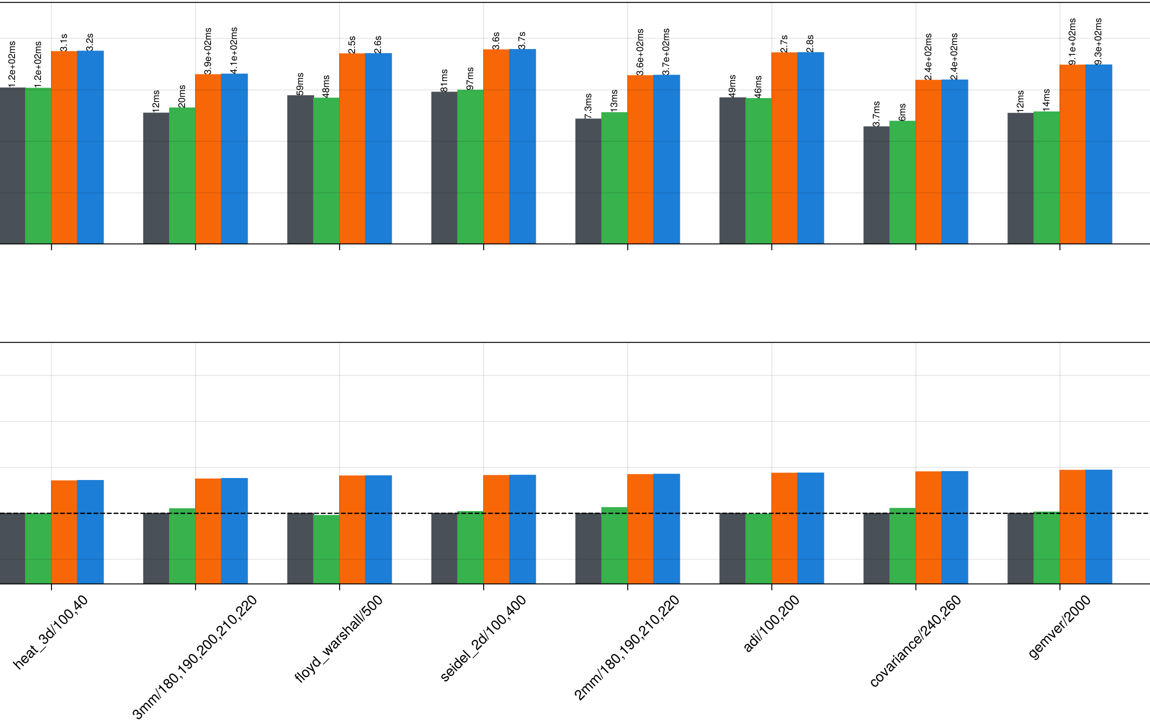

68s
66s
2.5s
68s
65s
66s
2.5e+02ms
66s
10s
8.2s
2.5s
11s
48s
49s
1.6s
50s
57s
62s
2.5s
64s
3.2s
3.4s
2.4e+02ms
3.6s
28s
30s
1.3s
32s
58s
64s
2s
66s
floyd_warshall/5600
covariance/2600,3000
floyd_warshall/2800
syr2k/2000,2600
nussinov/5500
covariance/1200,1400
symm/2000,2600
ludcmp/4000

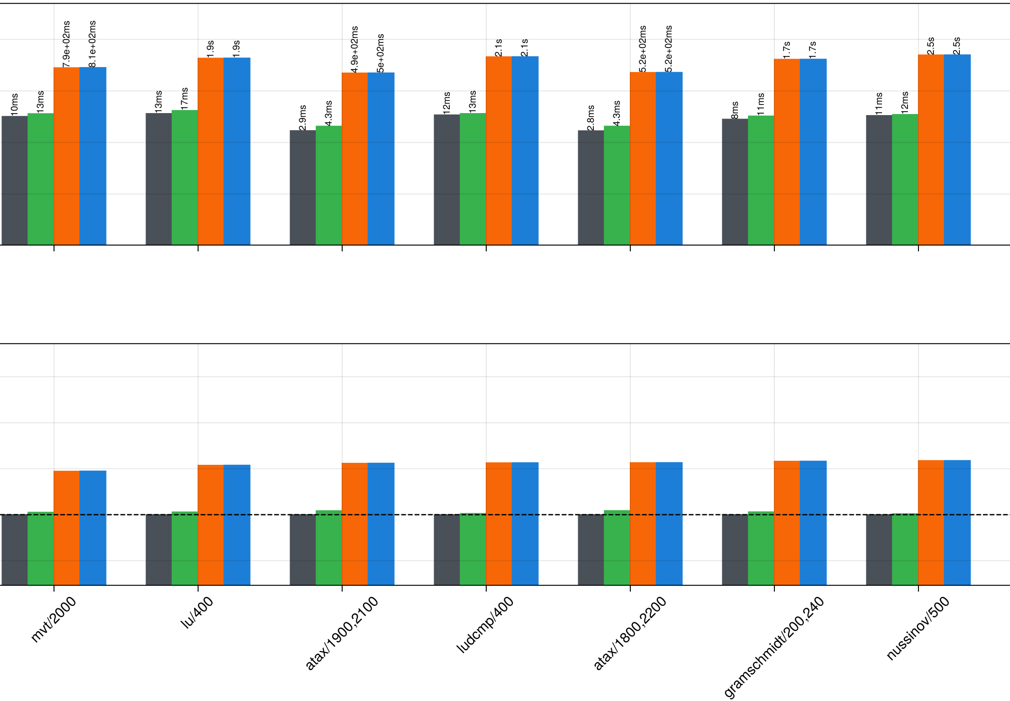

14s
13s
3s
16s
36s
42s
4e+02ms
43s
21s
25s
3.9e+02ms
25s
11s
11s
2.9s
14s
6.9s
6s
2.9s
8.9s
11s
11s
3.7s
15s
12s
14s
1.8s
16s
11s
11s
3.7s
15s
heat_3d/1000,200
3mm/1600,1800,2000,2200,2400
2mm/1600,1800,2200,2400
fdtd_2d/1000,2000,2600
adi/500,1000
jacobi_2d/1000,2800
syrk/2000,2600
seidel_2d/500,2000

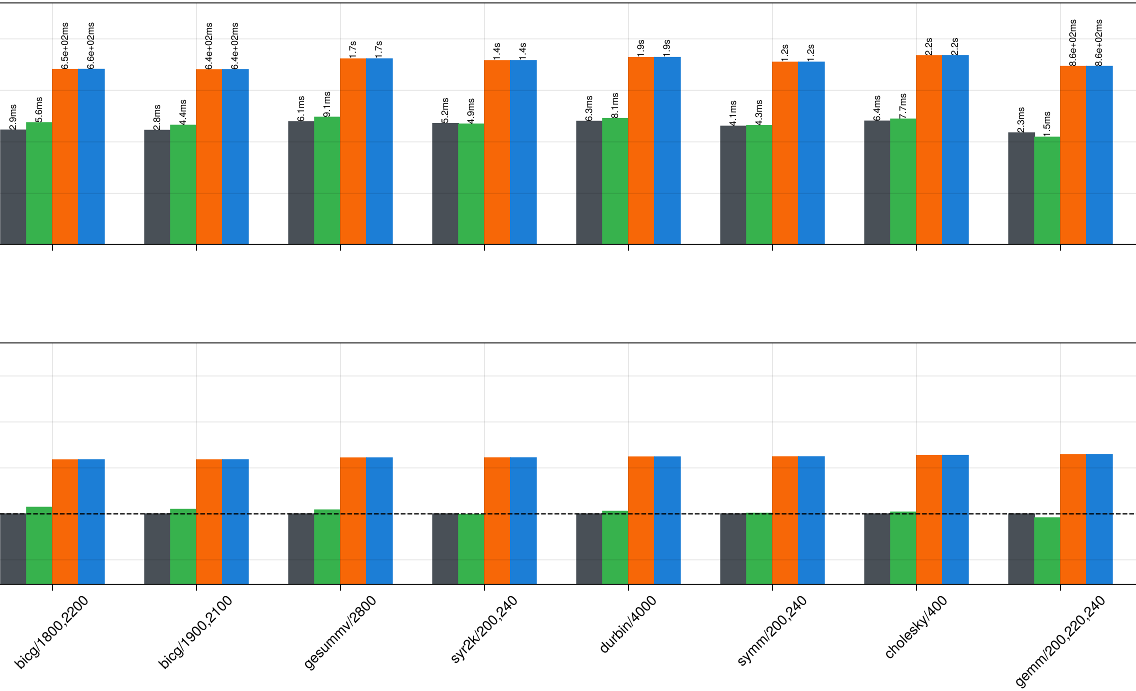

8.9s
11s
2.2s
13s
3.7s
4s
1.7s
5.6s
2.4s
2.3s
1.5s
3.8s
2.8s
2.8s
1.9s
4.7s
1.9s
3s
3.9e+02ms
3.3s
1.2s
1.9s
3.7e+02ms
2.3s
2.5s
2.8s
2s
4.8s
cholesky/4000
gramschmidt/1000,1200
syr2k/1000,1200
lu/2000
3mm/800,900,1000,1100,1200
2mm/800,900,1100,1200
ludcmp/2000

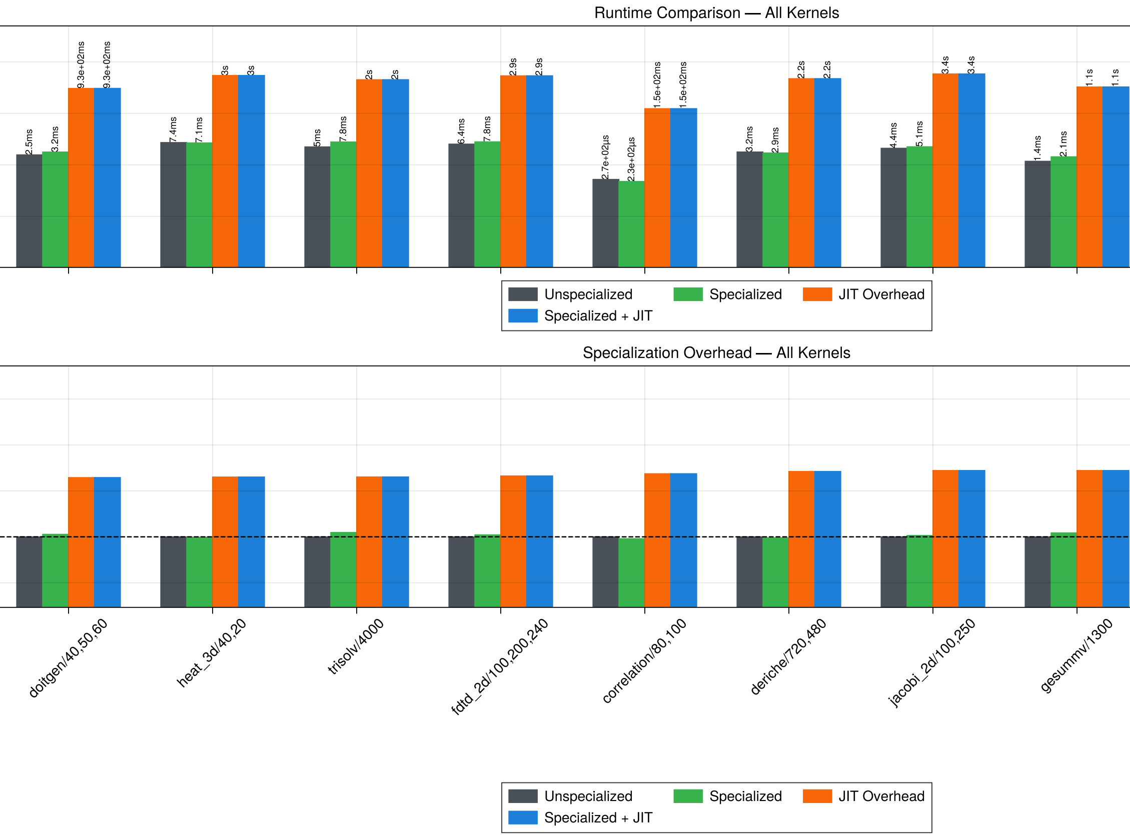

2.2s
3.3s
9.8e+02ms
4.2s
3.7s
4.3s
3s
7.3s
1s
1.1s
1.3s
2.3s
1.9s
2.5s
2.5s
5s
9.4e+02ms
9.8e+02ms
2.1s
3s
3.3e+02ms
2.4e+02ms
8.4e+02ms
1.1s
1.2s
1.3s
2.9s
4.3s
9.1e+02ms
1.2s
2.4s
3.6s
doitgen/220,250,270
heat_3d/500,120
symm/1000,1200
nussinov/2500
trmm/1000,1200
gemm/1000,1100,1200
fdtd_2d/500,1000,1200
cholesky/2000

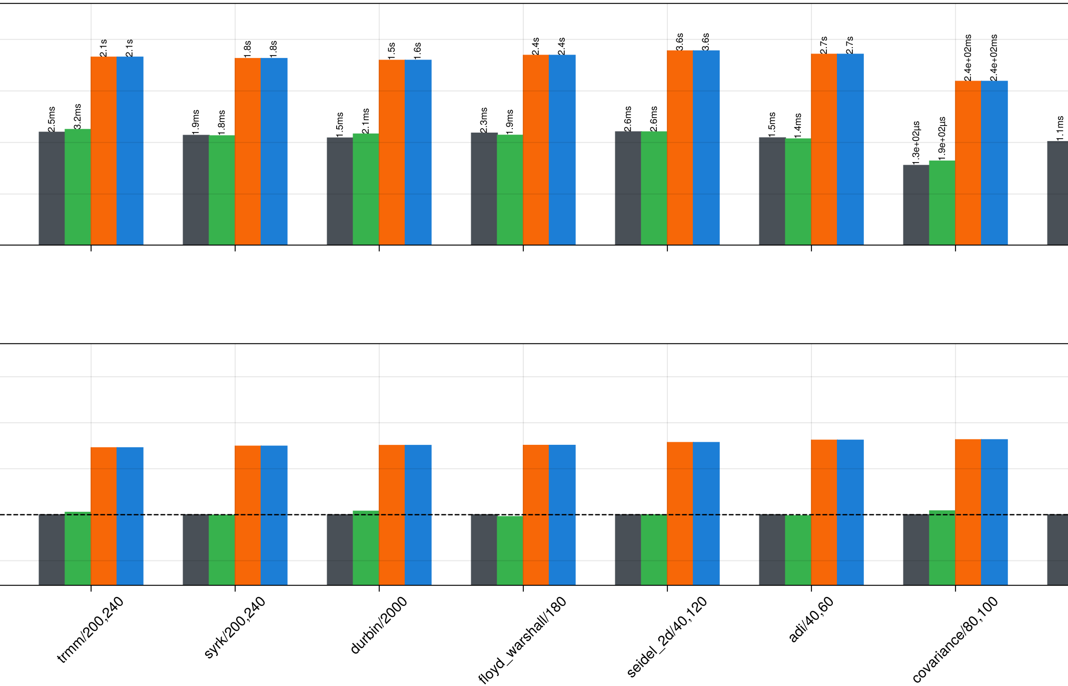

5.7e+02ms
5.7e+02ms
1.8s
2.4s
6.5e+02ms
7.5e+02ms
2.3s
3s
9.9e+02ms
1e+03ms
3.6s
4.6s
2.8e+02ms
3.8e+02ms
9.5e+02ms
1.3s
90ms
1.1e+02ms
6.8e+02ms
7.9e+02ms
91ms
1.1e+02ms
9.5e+02ms
1.1s
1.6e+02ms
1.8e+02ms
2.2s
2.4s
7.2ms
7.3ms
1.5e+02ms
1.6e+02ms
syrk/1000,1200
deriche/7680,4320
jacobi_2d/500,1300
doitgen/140,150,160
mvt/4000
gemver/4000
deriche/4096,2160
correlation/240,260

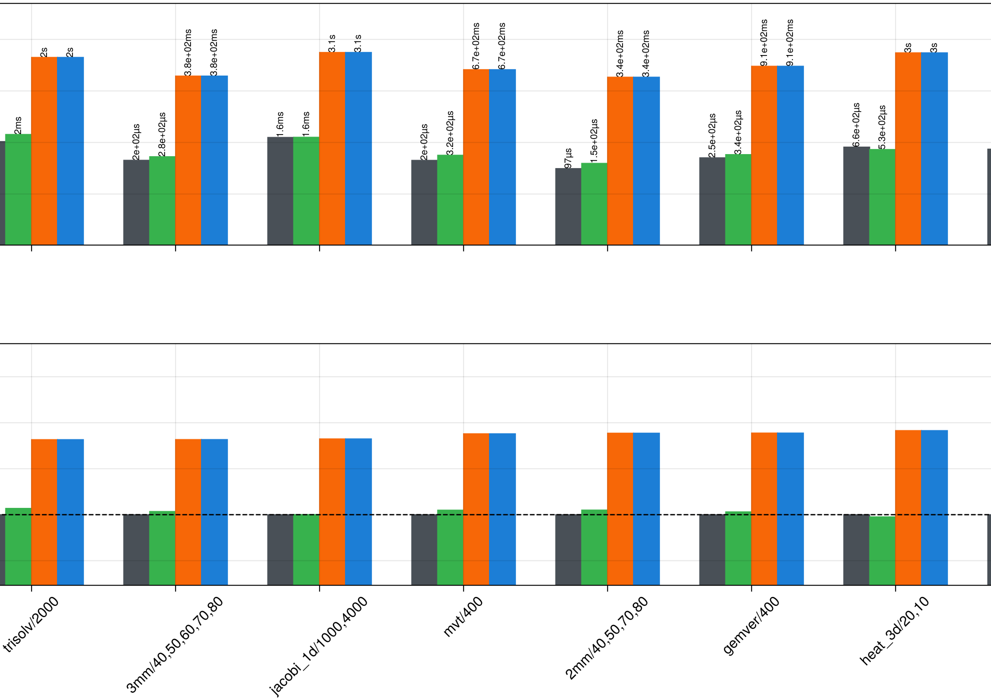

1.2e+02ms
1.2e+02ms
3.1s
3.2s
12ms
20ms
3.9e+02ms
4.1e+02ms
59ms
48ms
2.5s
2.6s
81ms
97ms
3.6s
3.7s
7.3ms
13ms
3.6e+02ms
3.7e+02ms
49ms
46ms
2.7s
2.8s
3.7ms
6ms
2.4e+02ms
2.4e+02ms
12ms
14ms
9.1e+02ms
9.3e+02ms
heat_3d/100,40
3mm/180,190,200,210,220
floyd_warshall/500
seidel_2d/100,400
2mm/180,190,210,220
adi/100,200
covariance/240,260
gemver/2000

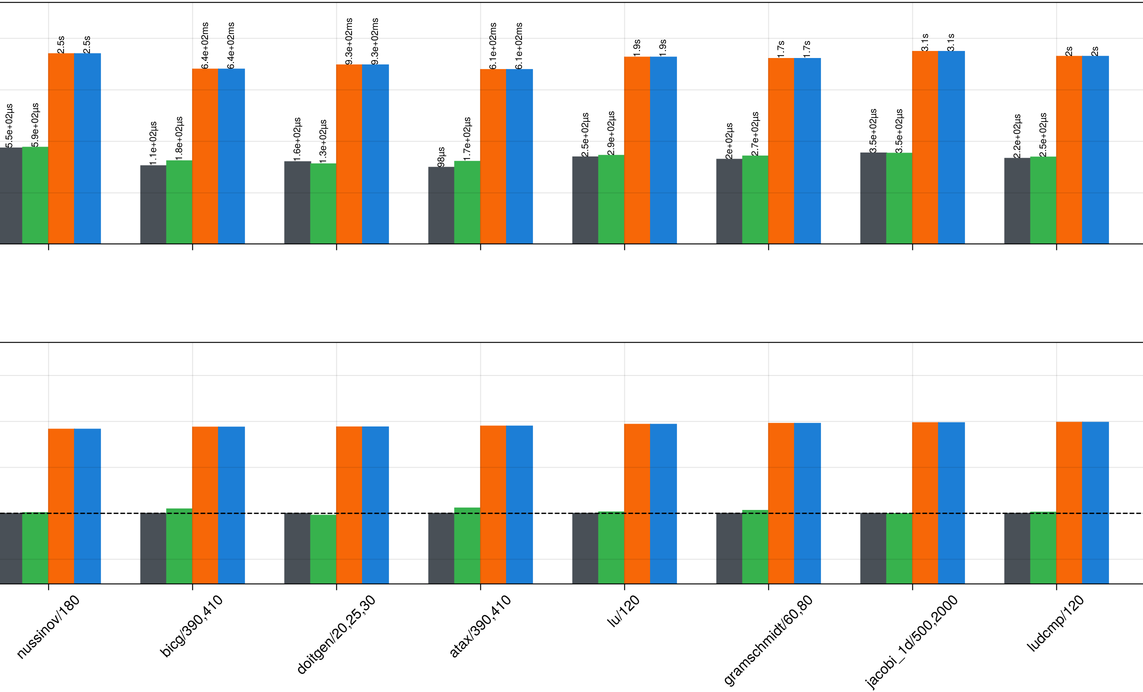

10ms
13ms
7.9e+02ms
8.1e+02ms
13ms
17ms
1.9s
1.9s
2.9ms
4.3ms
4.9e+02ms
5e+02ms
12ms
13ms
2.1s
2.1s
2.8ms
4.3ms
5.2e+02ms
5.2e+02ms
8ms
11ms
1.7s
1.7s
11ms
12ms
2.5s
2.5s
mvt/2000
lu/400
atax/1900,2100
ludcmp/400
atax/1800,2200
gramschmidt/200,240
nussinov/500

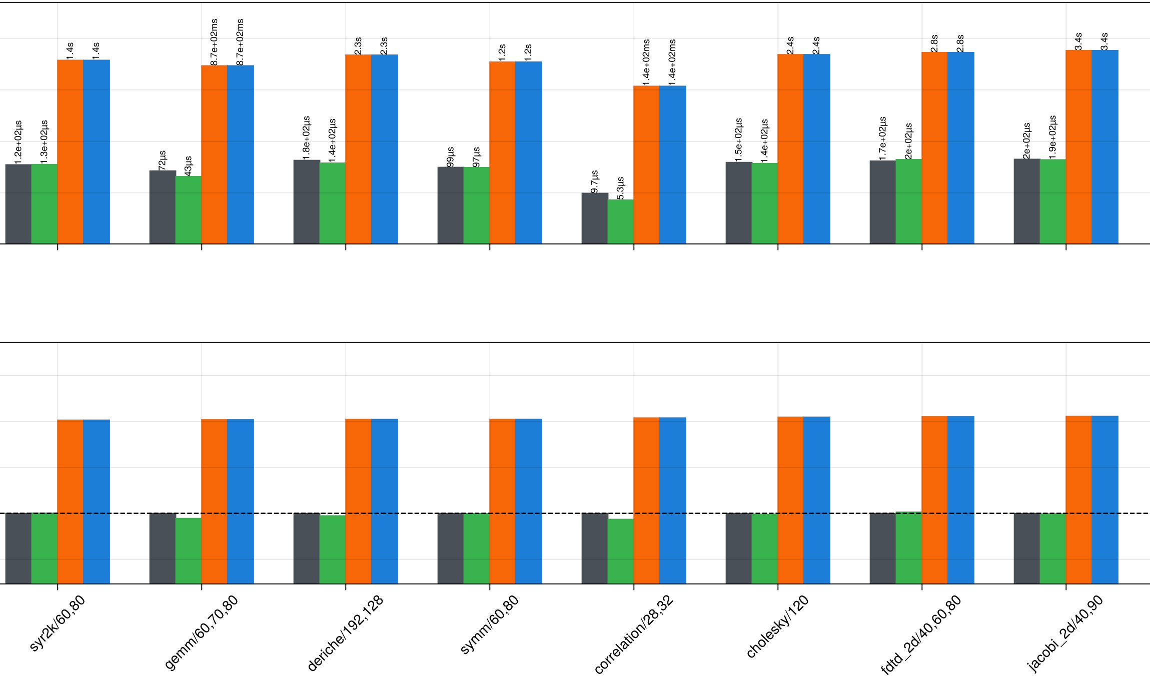

2.9ms
5.6ms
6.5e+02ms
6.6e+02ms
2.8ms
4.4ms
6.4e+02ms
6.4e+02ms
6.1ms
9.1ms
1.7s
1.7s
5.2ms
4.9ms
1.4s
1.4s
6.3ms
8.1ms
1.9s
1.9s
4.1ms
4.3ms
1.2s
1.2s
6.4ms
7.7ms
2.2s
2.2s
2.3ms
1.5ms
8.6e+02ms
8.6e+02ms
bicg/1800,2200
bicg/1900,2100
gesummv/2800
syr2k/200,240
durbin/4000
symm/200,240
cholesky/400
gemm/200,220,240

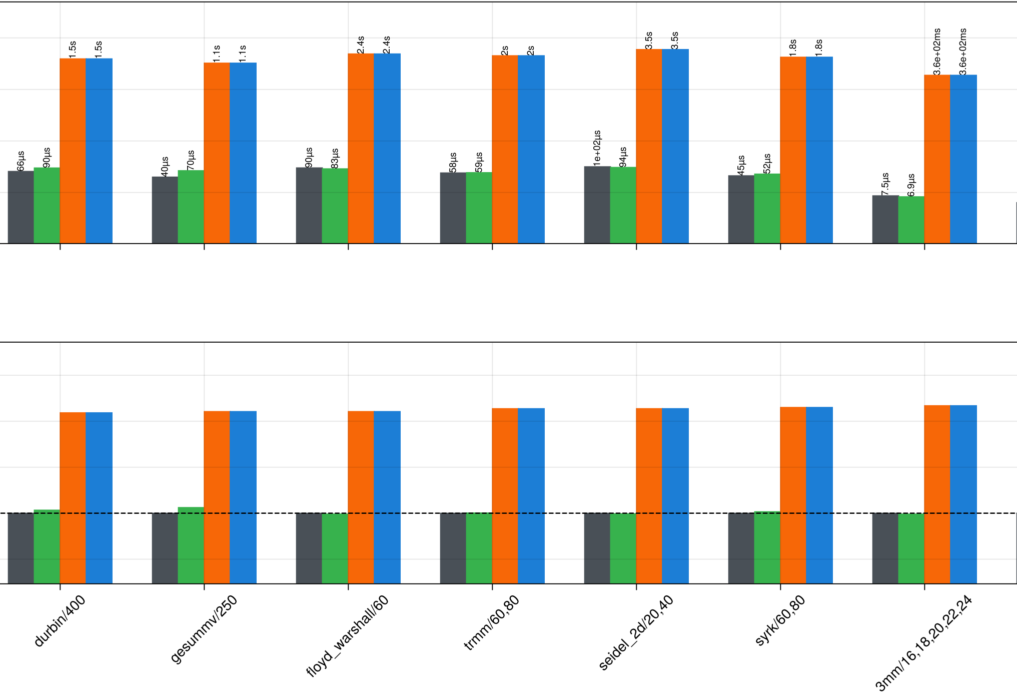

Runtime Comparison — All Kernels
2.5ms
3.2ms
9.3e+02ms
9.3e+02ms
7.4ms
7.1ms
3s
3s
5ms
7.8ms
2s
2s
6.4ms
7.8ms
2.9s
2.9s
2.7e+02µs
2.3e+02µs
1.5e+02ms
1.5e+02ms
3.2ms
2.9ms
2.2s
2.2s
4.4ms
5.1ms
3.4s
3.4s
1.4ms
2.1ms
1.1s
1.1s
Unspecialized
Specialized
JIT Overhead
Specialized + JIT
Specialization Overhead — All Kernels
doitgen/40,50,60
heat_3d/40,20
trisolv/4000
fdtd_2d/100,200,240
correlation/80,100
deriche/720,480
jacobi_2d/100,250
gesummv/1300
Unspecialized
Specialized
JIT Overhead
Specialized + JIT

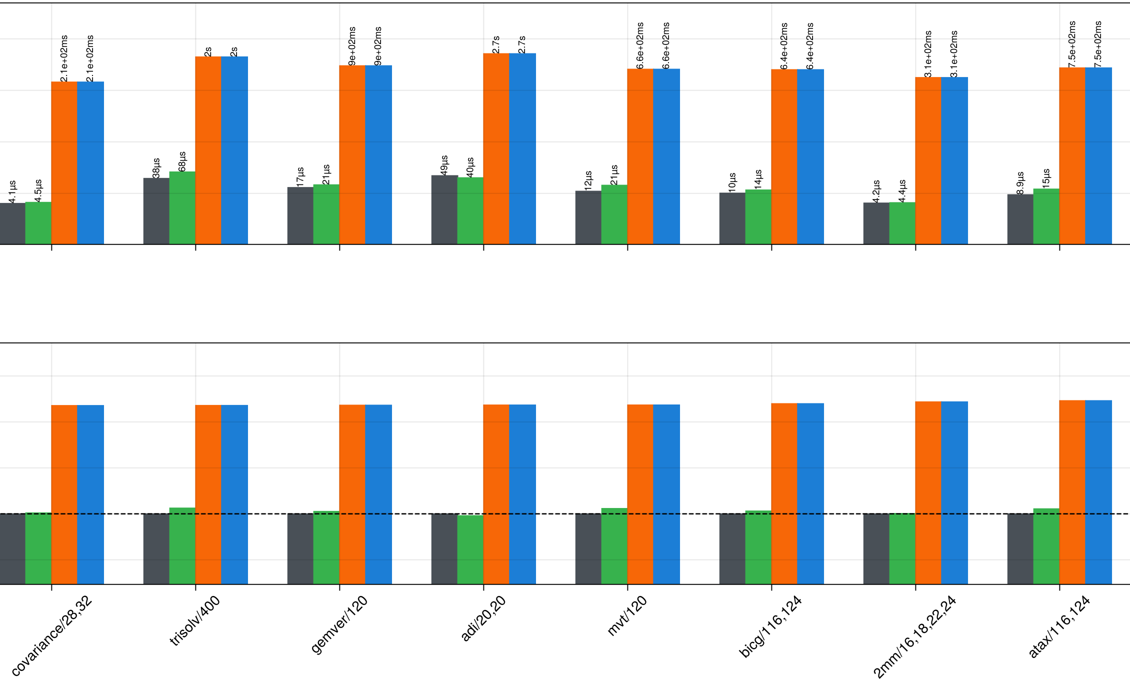

2.5ms
3.2ms
2.1s
2.1s
1.9ms
1.8ms
1.8s
1.8s
1.5ms
2.1ms
1.5s
1.6s
2.3ms
1.9ms
2.4s
2.4s
2.6ms
2.6ms
3.6s
3.6s
1.5ms
1.4ms
2.7s
2.7s
1.3e+02µs
1.9e+02µs
2.4e+02ms
2.4e+02ms
1.1ms
trmm/200,240
syrk/200,240
durbin/2000
floyd_warshall/180
seidel_2d/40,120
adi/40,60
covariance/80,100

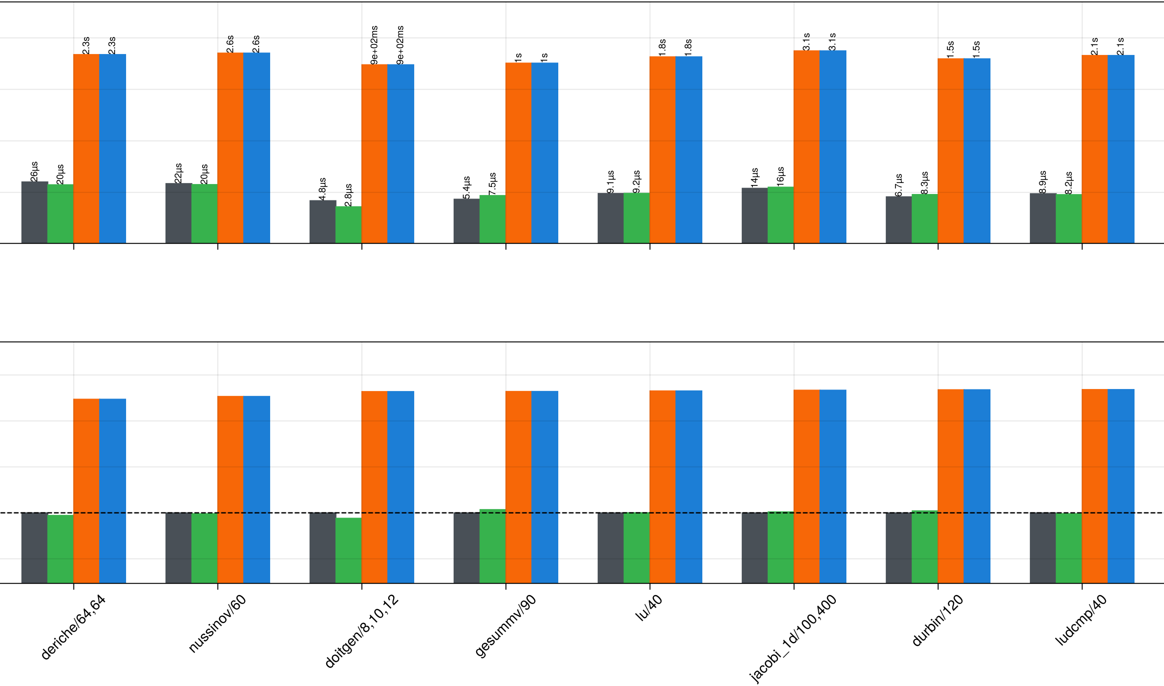

2ms
2s
2s
2e+02µs
2.8e+02µs
3.8e+02ms
3.8e+02ms
1.6ms
1.6ms
3.1s
3.1s
2e+02µs
3.2e+02µs
6.7e+02ms
6.7e+02ms
97µs
1.5e+02µs
3.4e+02ms
3.4e+02ms
2.5e+02µs
3.4e+02µs
9.1e+02ms
9.1e+02ms
6.6e+02µs
5.3e+02µs
3s
3s
trisolv/2000
3mm/40,50,60,70,80
jacobi_1d/1000,4000
mvt/400
2mm/40,50,70,80
gemver/400
heat_3d/20,10

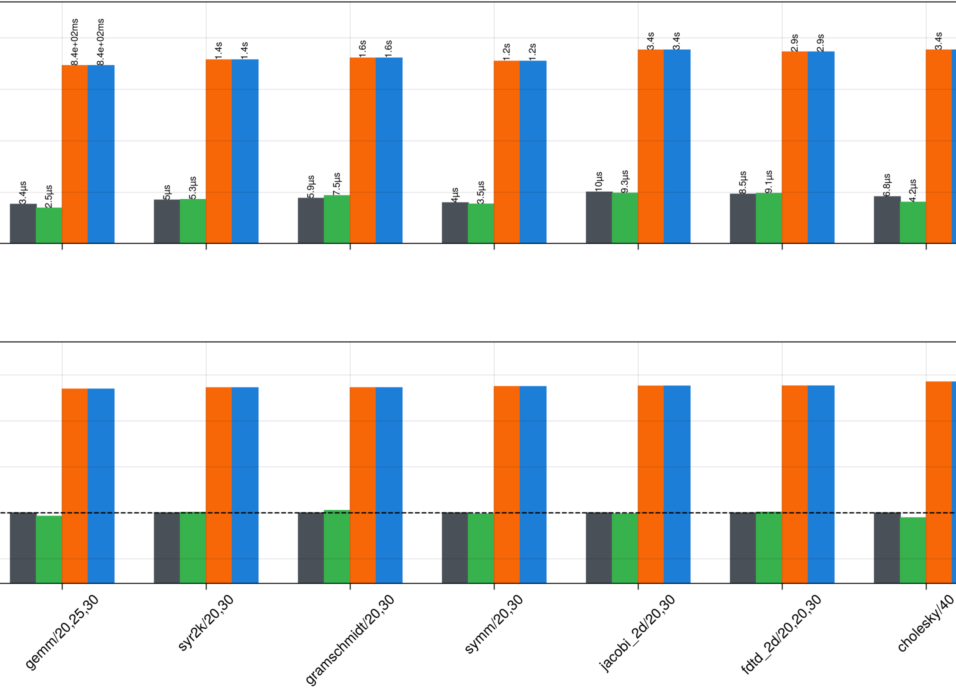

5.5e+02µs
5.9e+02µs
2.5s
2.5s
1.1e+02µs
1.8e+02µs
6.4e+02ms
6.4e+02ms
1.6e+02µs
1.3e+02µs
9.3e+02ms
9.3e+02ms
98µs
1.7e+02µs
6.1e+02ms
6.1e+02ms
2.5e+02µs
2.9e+02µs
1.9s
1.9s
2e+02µs
2.7e+02µs
1.7s
1.7s
3.5e+02µs
3.5e+02µs
3.1s
3.1s
2.2e+02µs
2.5e+02µs
2s
2s
nussinov/180
bicg/390,410
doitgen/20,25,30
atax/390,410
lu/120
gramschmidt/60,80
jacobi_1d/500,2000
ludcmp/120

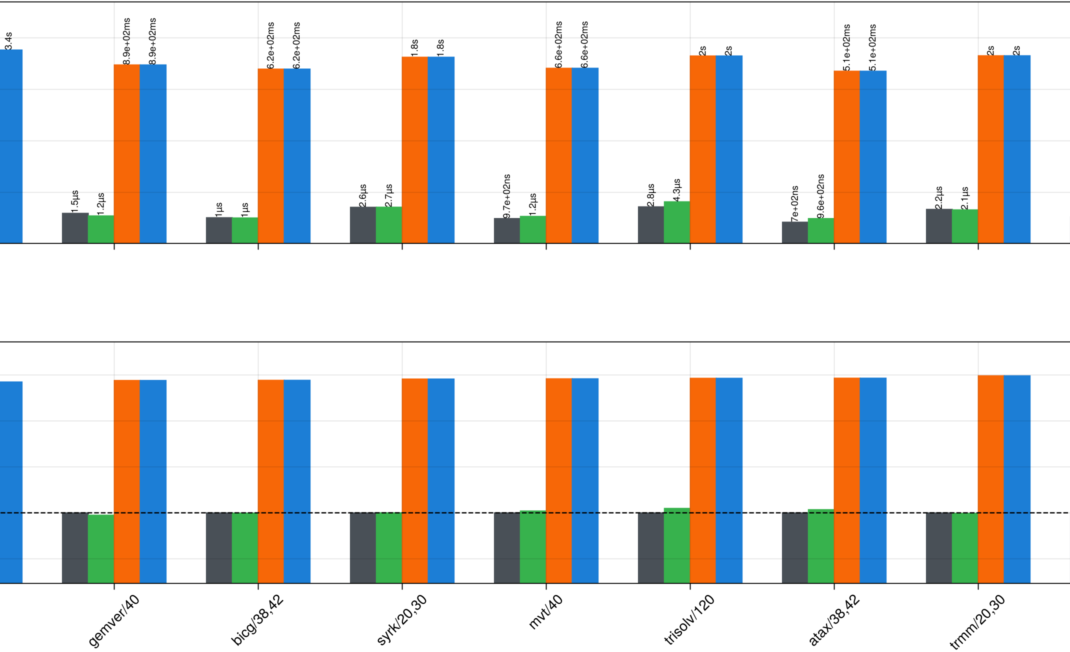

1.2e+02µs
1.3e+02µs
1.4s
1.4s
72µs
43µs
8.7e+02ms
8.7e+02ms
1.8e+02µs
1.4e+02µs
2.3s
2.3s
99µs
97µs
1.2s
1.2s
9.7µs
5.3µs
1.4e+02ms
1.4e+02ms
1.5e+02µs
1.4e+02µs
2.4s
2.4s
1.7e+02µs
2e+02µs
2.8s
2.8s
2e+02µs
1.9e+02µs
3.4s
3.4s
syr2k/60,80
gemm/60,70,80
deriche/192,128
symm/60,80
correlation/28,32
cholesky/120
fdtd_2d/40,60,80
jacobi_2d/40,90

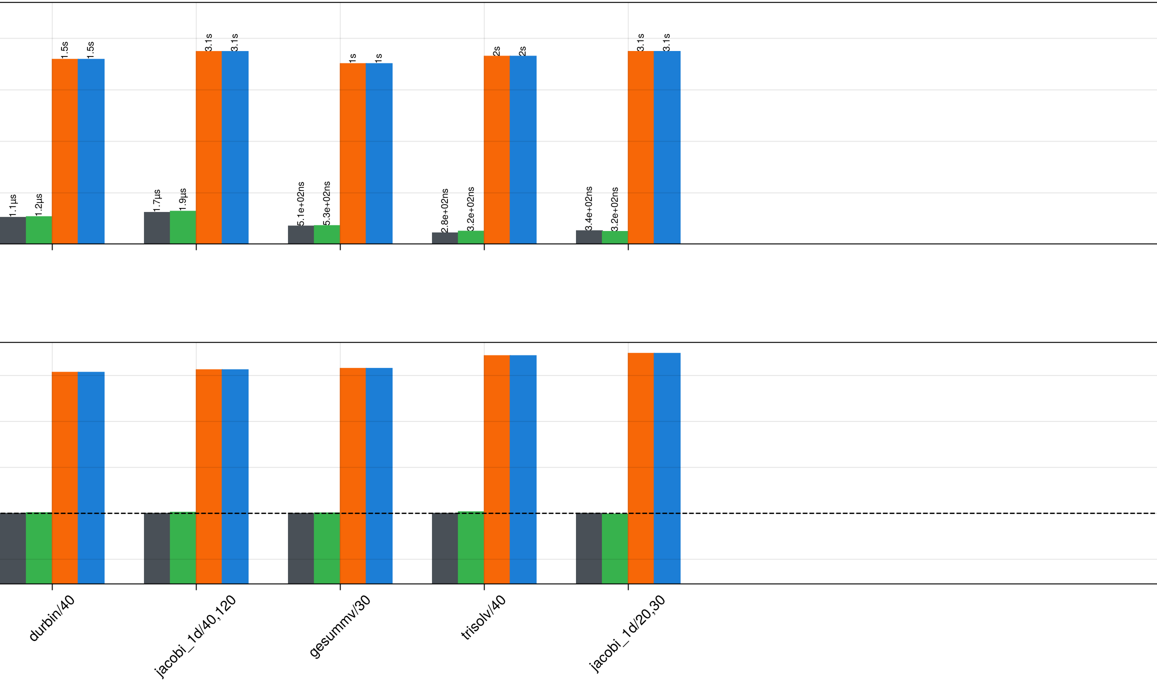

66µs
90µs
1.5s
1.5s
40µs
70µs
1.1s
1.1s
90µs
83µs
2.4s
2.4s
58µs
59µs
2s
2s
1e+02µs
94µs
3.5s
3.5s
45µs
52µs
1.8s
1.8s
7.5µs
6.9µs
3.6e+02ms
3.6e+02ms
durbin/400
gesummv/250
floyd_warshall/60
trmm/60,80
seidel_2d/20,40
syrk/60,80
3mm/16,18,20,22,24


4.1µs
4.5µs
2.1e+02ms
2.1e+02ms
38µs
68µs
2s
2s
17µs
21µs
9e+02ms
9e+02ms
49µs
40µs
2.7s
2.7s
12µs
21µs
6.6e+02ms
6.6e+02ms
10µs
14µs
6.4e+02ms
6.4e+02ms
4.2µs
4.4µs
3.1e+02ms
3.1e+02ms
8.9µs
15µs
7.5e+02ms
7.5e+02ms
covariance/28,32
trisolv/400
gemver/120
adi/20,20
mvt/120
bicg/116,124
2mm/16,18,22,24
atax/116,124


26µs
20µs
2.3s
2.3s
22µs
20µs
2.6s
2.6s
4.8µs
2.8µs
9e+02ms
9e+02ms
5.4µs
7.5µs
1s
1s
9.1µs
9.2µs
1.8s
1.8s
14µs
16µs
3.1s
3.1s
6.7µs
8.3µs
1.5s
1.5s
8.9µs
8.2µs
2.1s
2.1s
deriche/64,64
nussinov/60
doitgen/8,10,12
gesummv/90
lu/40
jacobi_1d/100,400
durbin/120
ludcmp/40


3.4µs
2.5µs
8.4e+02ms
8.4e+02ms
5µs
5.3µs
1.4s
1.4s
5.9µs
7.5µs
1.6s
1.6s
4µs
3.5µs
1.2s
1.2s
10µs
9.3µs
3.4s
3.4s
8.5µs
9.1µs
2.9s
2.9s
6.8µs
4.2µs
3.4s
gemm/20,25,30
syr2k/20,30
gramschmidt/20,30
symm/20,30
jacobi_2d/20,30
fdtd_2d/20,20,30
cholesky/40


3.4s
1.5µs
1.2µs
8.9e+02ms
8.9e+02ms
1µs
1µs
6.2e+02ms
6.2e+02ms
2.6µs
2.7µs
1.8s
1.8s
9.7e+02ns
1.2µs
6.6e+02ms
6.6e+02ms
2.8µs
4.3µs
2s
2s
7e+02ns
9.6e+02ns
5.1e+02ms
5.1e+02ms
2.2µs
2.1µs
2s
2s
gemver/40
bicg/38,42
syrk/20,30
mvt/40
trisolv/120
atax/38,42
trmm/20,30


1.1µs
1.2µs
1.5s
1.5s
1.7µs
1.9µs
3.1s
3.1s
5.1e+02ns
5.3e+02ns
1s
1s
2.8e+02ns
3.2e+02ns
2s
2s
3.4e+02ns
3.2e+02ns
3.1s
3.1s
durbin/40
jacobi_1d/40,120
gesummv/30
trisolv/40
jacobi_1d/20,30